

HASS DIGITAL CHAMPIONS PROGRAM - INTRO PRESENTATION

DR. MARTIN SCHWEINBERGER

INTRODUCTION - DR MARTIN SCHWEINBERGER

- UQ POSTDOCTORAL RESEARCH FELLOW IN RESEARCH TECHNOLOGIES (CONSULTATION ON DATA VISUALIZATION AND (QUANTITATIVE) DATA ANALYSIS)
- MA PHILOSOPHY, PSYCHOLOGY, AND ENGLISH PHILOLOGY (KASSEL UNIVERSITY/ NATIONAL UNIVERSITY OF IRELAND, GALWAY)
- PHD IN ENGLISH LINGUISTICS (HAMBURG UNIVERSITY)
- BACKGROUND IN CORPUS LINGUISTICS, TEXT ANALYSIS, QUANTITATIVE DATA ANALYSIS, AND NLP (MOSTLY R AND A TINY BIT OF PYTHON)
- RESEARCH INTERESTS
 - SOCIO- AND PSYCHOLINGUISTICS
 - LANGUAGE VARIATION AND CHANGE
 - L1 AND L2 ACQUISITION
 - COMPUTATIONAL ANALYSIS OF SPEECH (PRAAT)
- LAST POSITION: COMPUTER SCIENCE (AUTOMATED SPEECH EVALUATION) (BEFORE INTERIM PROFESSOR FOR ENGLISH LINGUISTICS, HAMBURG UNIVERSITY)



CURRENT PROJECT

ACQUISITION, VARIATION, AND DIACHRONIC CHANGE IN AND ACROSS ENGLISH AMPLIFIER SYSTEMS

- (1) A. I FEEL ASSURED THAT I LOOKED **VERY TERRIFIC**. (COHA:FIC:1846)
B. PAMELA WAS **REALLY COOL** IN THE MIDST OF ALL THIS. (COHA:MAG:1993)
C. THAT IS **SO AWESOME**. (COHA: FIC:2007)
- HOW ARE AMPLIFIERS ACQUIRED?
 - HOW DO AMPLIFIERS VARY WITHIN AND ACROSS VARIETIES OF ENGLISH?
(IMPACT OF SOCIAL, LINGUISTIC, AND PSYCHOLOGICAL FACTORS?)
 - HOW DOES AMPLIFICATION CHANGE OVER TIME?

CURRENT PROJECT

ACQUISITION, VARIATION, AND DIACHRONIC CHANGE IN AND ACROSS ENGLISH AMPLIFIER SYSTEMS

DATA (VARIOUS CORPORA)

- CHILDES
- INTERNATIONAL CORPUS OF ENGLISH (ICE)
- CORPUS OF HISTORICAL AMERICAN ENGLISH (COHA)

DATA PROCESSING (ALL IN R: VERSATILE, REPRODUCIBLE)

- PART-OF-SPEECH ANNOTATION
- CONCORDANCING
- DATA CLEANING AND TABULARIZATION

CURRENT PROJECT

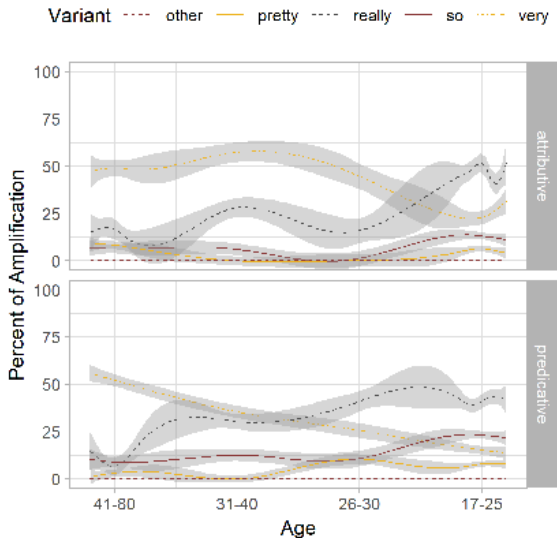
ACQUISITION, VARIATION, AND DIACHRONIC CHANGE IN AND ACROSS ENGLISH AMPLIFIER SYSTEMS

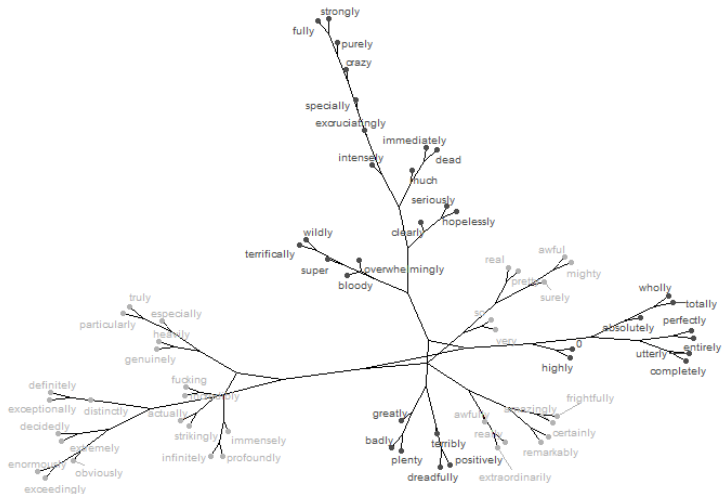
VISUALIZING THE DATA (MOSTLY **GGPLOT2** PACKAGE IN **R**)

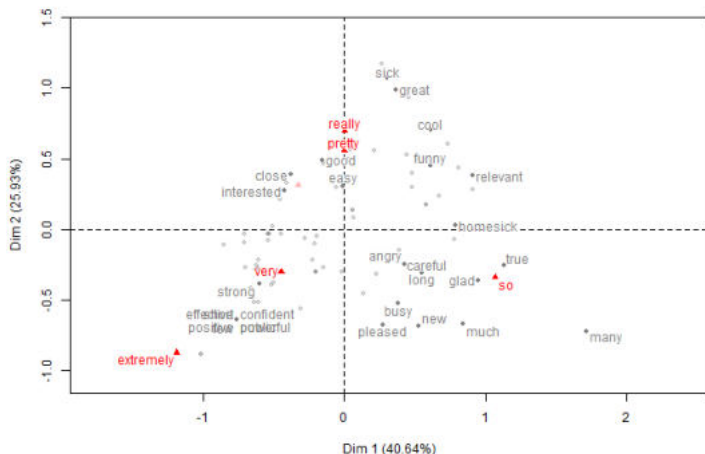
- LINE GRAPHS
- DENDROGRAMS
- BILOTS, ETC.

DATA ANALYSIS (STATISTICS)

- MIXED-EFFECTS LOGISTIC REGRESSIONS
- RANDOM FORESTS
- VECTOR SPACE MODELS
- COVARYING COLLEXEME ANALYSES
- CORRESPONDENCE AND CLUSTER ANALYSES

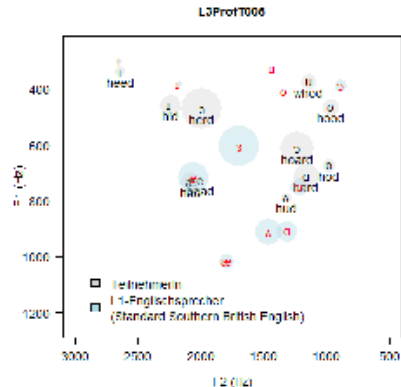






VOWELCHARTPROJECT

- ANALYZING VOWELS (FORMANTS, LENGTH) OF LANGUAGE LEARNERS AND COMPARING THEM TO L1-SPEAKERS (ONLY AME AND BrE SO FAR)
- ANALYSIS OF HOW/WHERE LEARNERS DIFFER FROM L1-SPEAKERS (WHERE ARE PROBLEMS, WHERE NOT?)
- SOME DATA ALREADY COLLECTED FOR ENGLISH, GERMAN, (SPANISH, AND RUSSIAN)
- ADDITIONAL ONLINE QUESTIONNAIRE ABOUT LEARNERS' LANGUAGE USE, THEIR LINGUISTIC BACKGROUND, STAYS ABROAD, EDUCATION HISTORY, MOTIVATIONAL ASPECTS, AND MUSICAL TRAINING TO FIND FACTORS THAT CONTRIBUTE TO NATIVE-LIKE PRONUNCIATION
- FEEDBACK TO/FOR LEARNERS: PERSONALIZED VOWEL CHARTS (OBJECTIVE FEEDBACK BASED ON SPEECH ANALYSIS)



HASS DIGITAL CHAMPIONS PROGRAM

WHAT ARE THE **key challenges** THAT YOU AND YOUR COMMUNITY ARE FACING?

- QUANTITATIVE TURN: (ADVANCED) STATISTICAL METHODS BECOME MORE AND MORE MAINSTREAM
- INCREASED INTEREST IN COMPUTER BASED ANALYSES OF LANGUAGE DATA (TEXT ANALYSIS, STATISTICS, DATA ANALYSIS, VISUALIZATION)
- SYSTEMATIC TRAINING IS STILL LACKING/WANTING
- LIMITED INTERACTION BETWEEN LINGUISTICS (HUMANITIES) AND COMPUTATIONAL LINGUISTIC (COMPUTER SCIENCE) ALTHOUGH LINGUISTICS, IN PARTICULAR, (BUT ALSO COMP. LING.) COULD/WOULD REALLY PROFIT FROM INTERACTION AND JOINED PROGRAMS!

HASS DIGITAL CHAMPIONS PROGRAM

WHAT DO YOU HOPE TO **achieve in the next 12 months** AND BEYOND

- TEXTBOOK ON TEXT ANALYSIS (EDINBURGH UNIVERSITY PRESS)
→ ACCOMPANYING ONLINE MATERIALS
- FAMILIARIZE STAFF AND JUNIOR RESEARCHERS WITH QUANTITATIVE AND COMPUTATIONAL METHODS AND SHOW THEM THAT THEY ARE REALLY USEFUL!
- BRIDGE THE GAP BETWEEN LINGUISTICS/HUMANITIES AND COMPUTATIONAL LINGUISTICS/COMPUTER SCIENCE